

Intelligent Candidate Ranking System

India Runs — Data & AI Challenge | Hack2Skill 2026

100K

Candidates Processed

0.9603

Top Score Achieved

5-Signal

Scoring Architecture

The Problem

Traditional ATS Fails

- ✗ Keyword filters miss real talent
- ✗ Scores a Marketing Manager #1 if she lists enough AI terms
- ✗ Ignores candidate availability & engagement
- ✗ No understanding of career context
- ✗ Cannot detect consulting-only profiles

Our Approach

- ✓ Multi-signal scoring across 5 dimensions
- ✓ Career text analysis (not just skill tags)
- ✓ Behavioral signals: recency, response rate
- ✓ Hard disqualifiers from JD applied directly
- ✓ Processes 100K candidates in ~30 seconds

"The right answer is NOT to find candidates whose skills section contains the most AI keywords. That's a trap." — JD itself

Scoring Architecture

30%

Skills Match

AI/ML core skills weighted by proficiency + endorsements + platform assessments

25%

Career Relevance

Title match, product company experience, career text scanning for key concepts

10%

Experience Years

Sweet spot: 5-9 years. Penalizes under/over-experienced candidates

25%

Behavioral Signals

Platform recency, open-to-work, recruiter response rate, notice period, GitHub

10%

Profile Quality

Completeness score, verified credentials, education tier, willing to relocate

$$\text{Final Score} = 0.30 \times \text{Skills} + 0.25 \times \text{Career} + 0.10 \times \text{Experience} + 0.25 \times \text{Behavioral} + 0.10 \times \text{Quality}$$

Reading Between the Lines

What the JD says vs what it means

What JD SAYS

- 5-9 years experience required
- Embeddings, vector DBs, Python
- LLM fine-tuning preferred
- Hybrid search knowledge
- Pune / Noida location

What JD MEANS

- Judgment matters more than years
- Production deployment matters
- Shipper > researcher mindset
- Never worked at consulting firm only
- Active on platform, short notice period

Hard DISQUALIFIERS

- TCS / Infosys / Wipro only career
- Marketing / HR / non-tech titles
- Inactive > 1 year on platform
- AI keywords, zero ML context
- Under 12-month LLM-only projects

The Behavioral Layer — Why It Matters

A perfect-on-paper candidate who hasn't logged in for 8 months is useless to a recruiter.

Last Active Date (35%)

Primary availability signal — inactive > 6mo
= heavy downweight

Open to Work Flag (15%)

Explicit intent signal on the Redrob platform

Recruiter Response Rate (20%)

Shows candidate actually responds —
critical for recruiters

Interview Completion (10%)

Shows reliability — will they actually show
up?

Notice Period (10%)

JD prefers sub-30 days; 30+ is penalized
progressively

GitHub Activity Score (10%)

External validation of technical work —
open-source signal

Results & Top Candidates

100,000

Total Candidates

0.9603

Highest Score

0.8448

Score at Rank 100

7.1 yrs

Avg Exp (Top 10)

Top 5 Ranked Candidates

Rank	Candidate ID	Title	Exp	AI Skills	Score
1	CAND_0018499	Sr. ML Engineer	7.2 yrs	12	0.9603
2	CAND_0077337	Staff ML Engineer	7.0 yrs	11	0.9455
3	CAND_0081846	Lead AI Engineer	6.7 yrs	13	0.9317
4	CAND_0050876	Applied ML Engineer	6.0 yrs	12	0.9314
5	CAND_0079387	AI Engineer	6.9 yrs	10	0.9301

Why This Beats Keyword Matching

Scenario 1: Marketing Manager with 15 AI keywords

✗ **Keyword:** Ranks HIGH — many keyword matches

✓ **Our System:** Ranks LOW — bad title penalty + < 3 AI skills in context → 60% score penalty

Scenario 2: Backend Engineer at Mindtree who built Kafka + Spark pipelines with ML model integration

✗ **Keyword:** Ranks LOW — title doesn't match

✓ **Our System:** Ranks HIGH — career text matches 'production, recommendation, data engineering'

Scenario 3: Senior ML Engineer, perfect skills, last active 14 months ago

✗ **Keyword:** Ranks HIGH — great keyword match

✓ **Our System:** Ranks DOWN — inactive > 1yr → 40% downweight

Tech Stack & Submission

Tech Stack

- ▶ Python 3.10+ — core language
- ▶ pandas / numpy — data handling
- ▶ JSON / CSV — input-output formats
- ▶ Pure multi-signal scoring — no heavy ML
- ▶ Runs 100K candidates in ~30 seconds
- ▶ No GPU required — laptop-friendly

Submission Structure

-  ranker.py — main scoring system
-  submission.csv — top 100 ranked
-  requirements.txt — dependencies
-  approach.pdf — this presentation
-  README.md — setup & methodology



Submission validated with official `validate_submission.py` — PASSED

Thank You

Making Hiring Smarter — One Signal at a Time

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